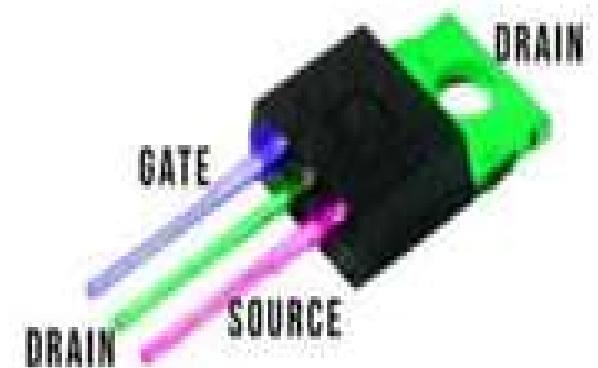


MOSFET

Metal Oxide Semiconductor
Field Effect Transistor



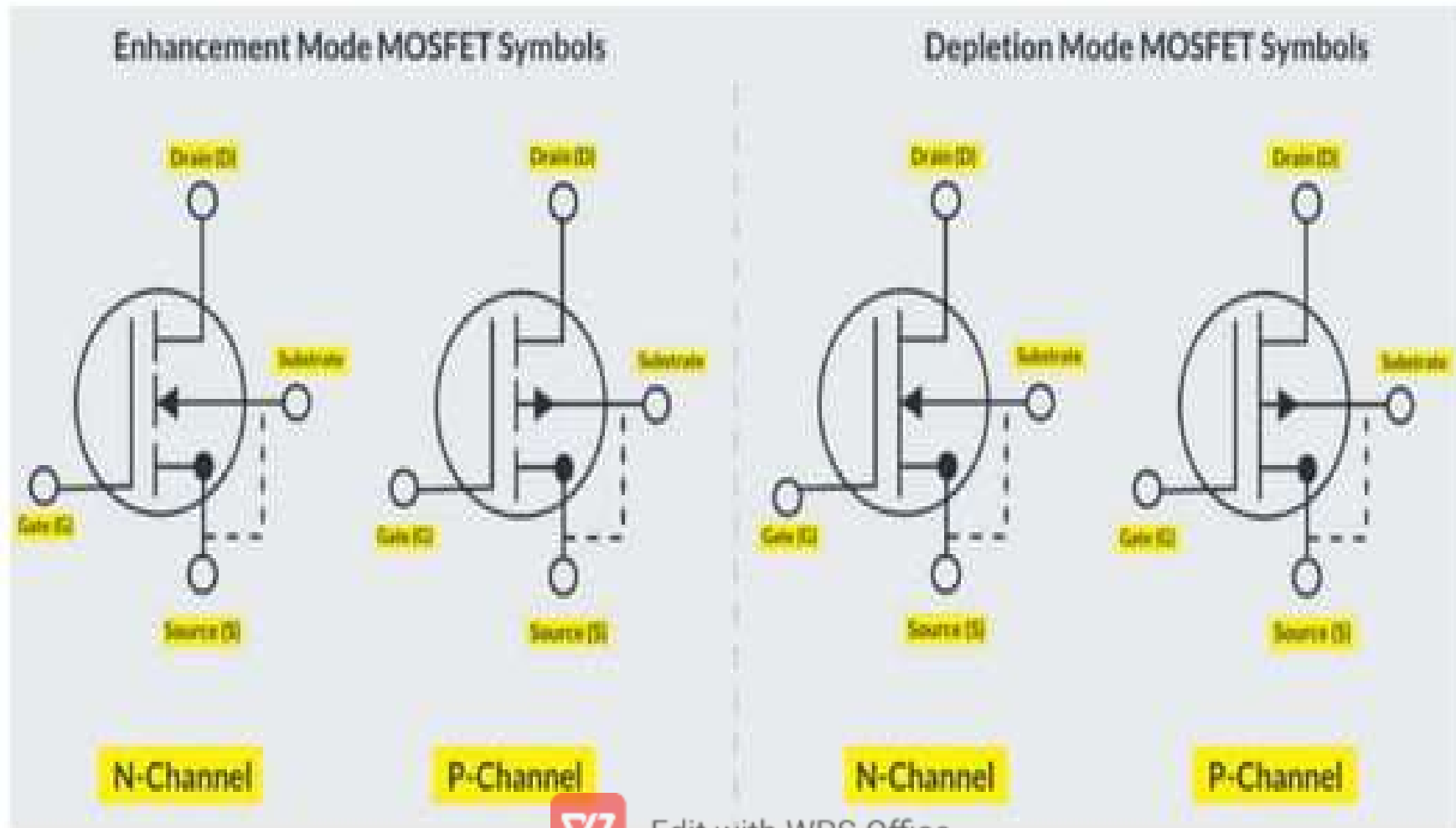
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Introduction

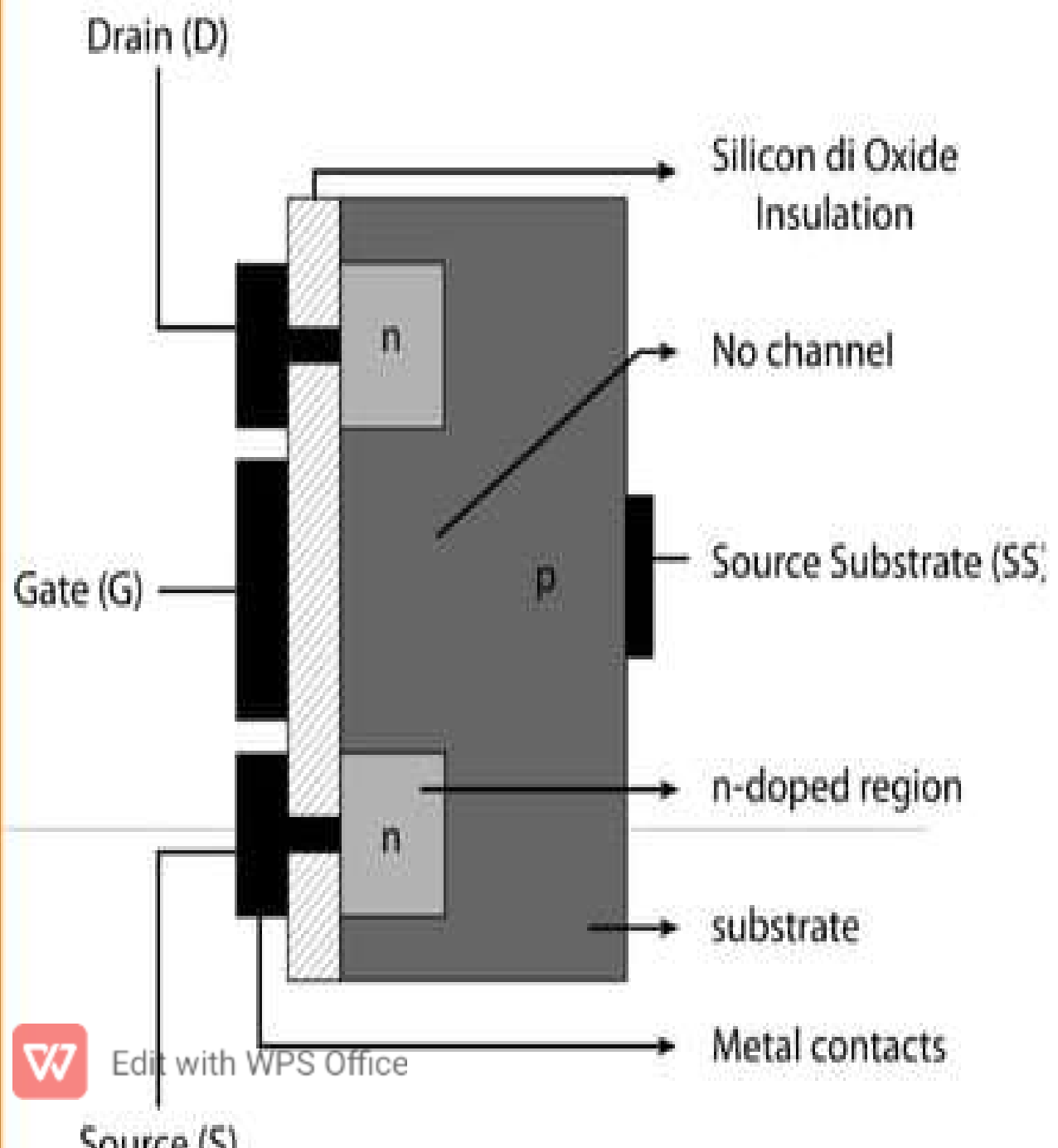
- MOSFETs are most common type of Field Effect Transistor
- MOSFETs are used in switching or amplifying electronic signals
- MOSFETs are voltage-controlled device
- MOSFETs are three terminal device with Source (S), Drain (D) and Gate (G) terminals.
- MOSFETs have very high switching speed
- There are 2 classes of MOSFETs : Enhancement and Depletion type
- These 2 classes are further divided as N-channel and P-channel



Types of MOSFET



N-channel Enhancement MOSFET

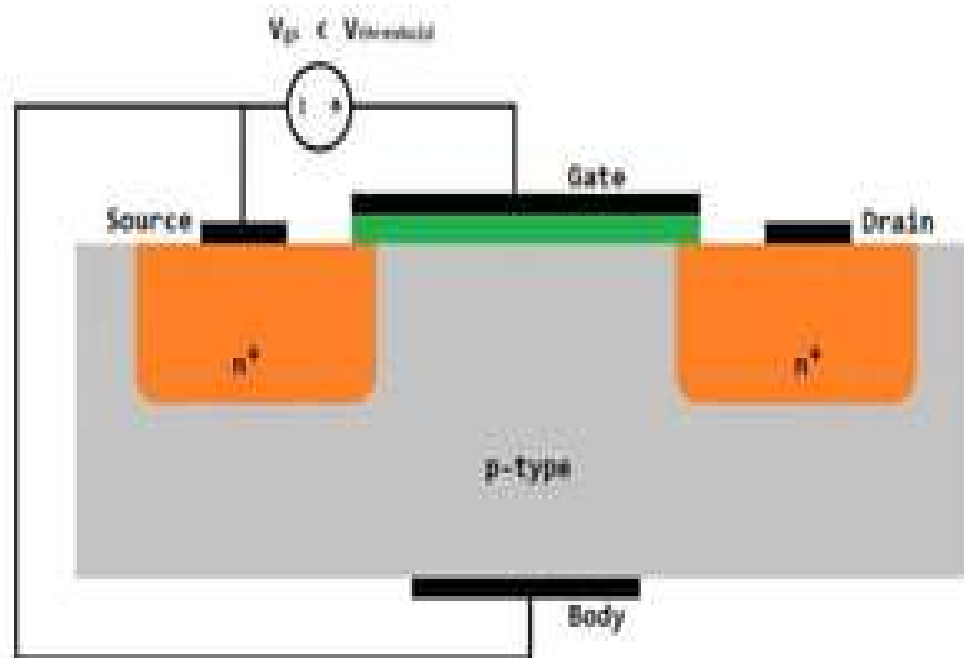


N-channel Enhancement MOSFET

- It has P-type semiconductor material as body/substrate
- Two N-type wells are diffused on the body/substrate
- Two PN junctions are formed creating 2 depletion regions
- A metal contact is created at the bottom and a terminal is taken out known as body/ substrate terminal
- Same metal contacts are created on the top of N-type making source and drain terminals
- There is a layer of Silicon Dioxide in between the 2 N-type well which act as a dielectric
- A metal contact is created on the top of SiO₂ layer and a terminal is taken out called Gate
- Generally Source and body terminal are internally connected and is grounded

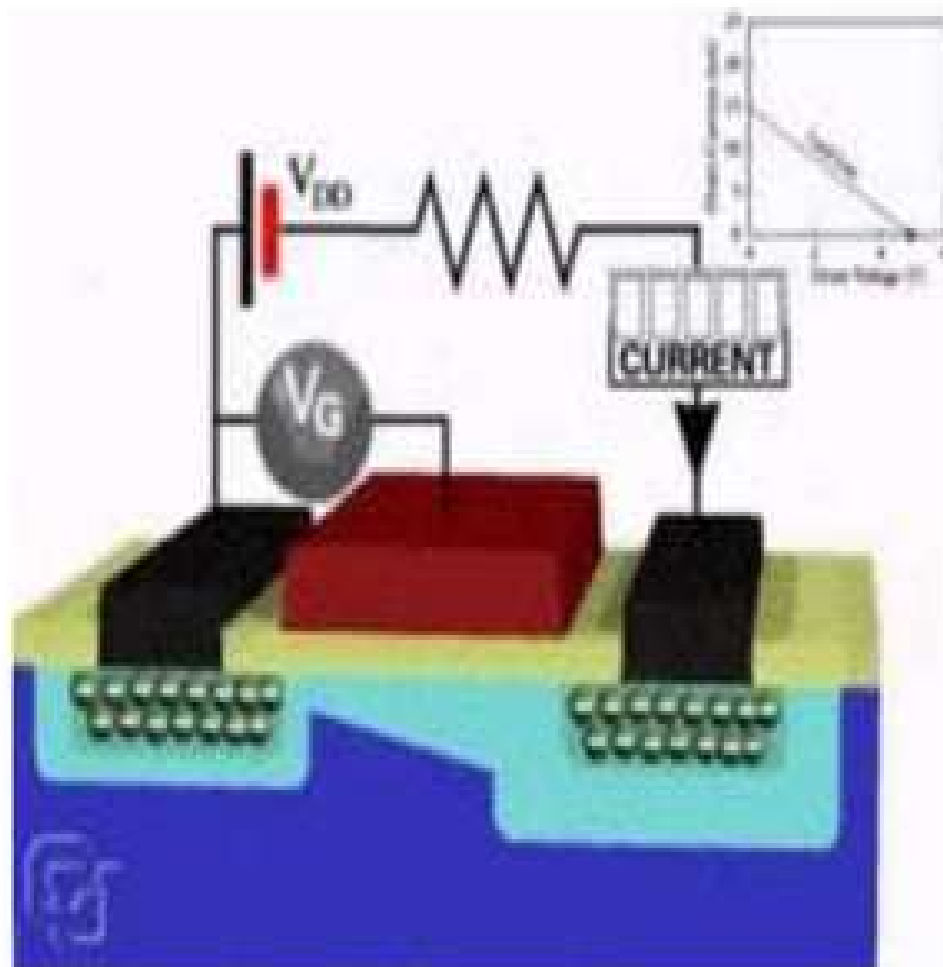


N-channel Enhancement MOSFET



- Apply V_{GS} between gate and source terminal keeping gate at high potential
- Electrons will push towards the gate terminal and accumulate there
- On increasing V_{GS} electron hole pairs are formed pushing holes below and accumulating more electrons near Gate
- When V_{GS} reaches a threshold voltage, an n type channel is formed



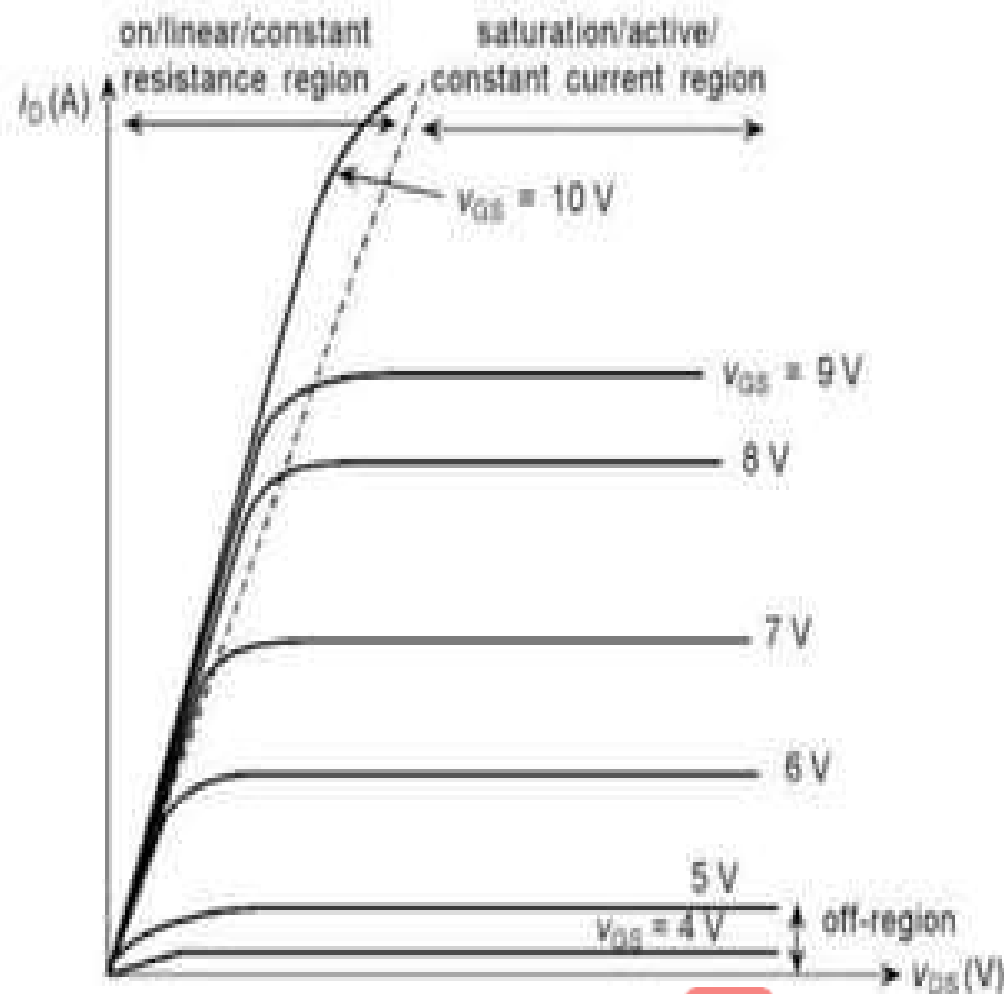


N-channel Enhancement MOSFET

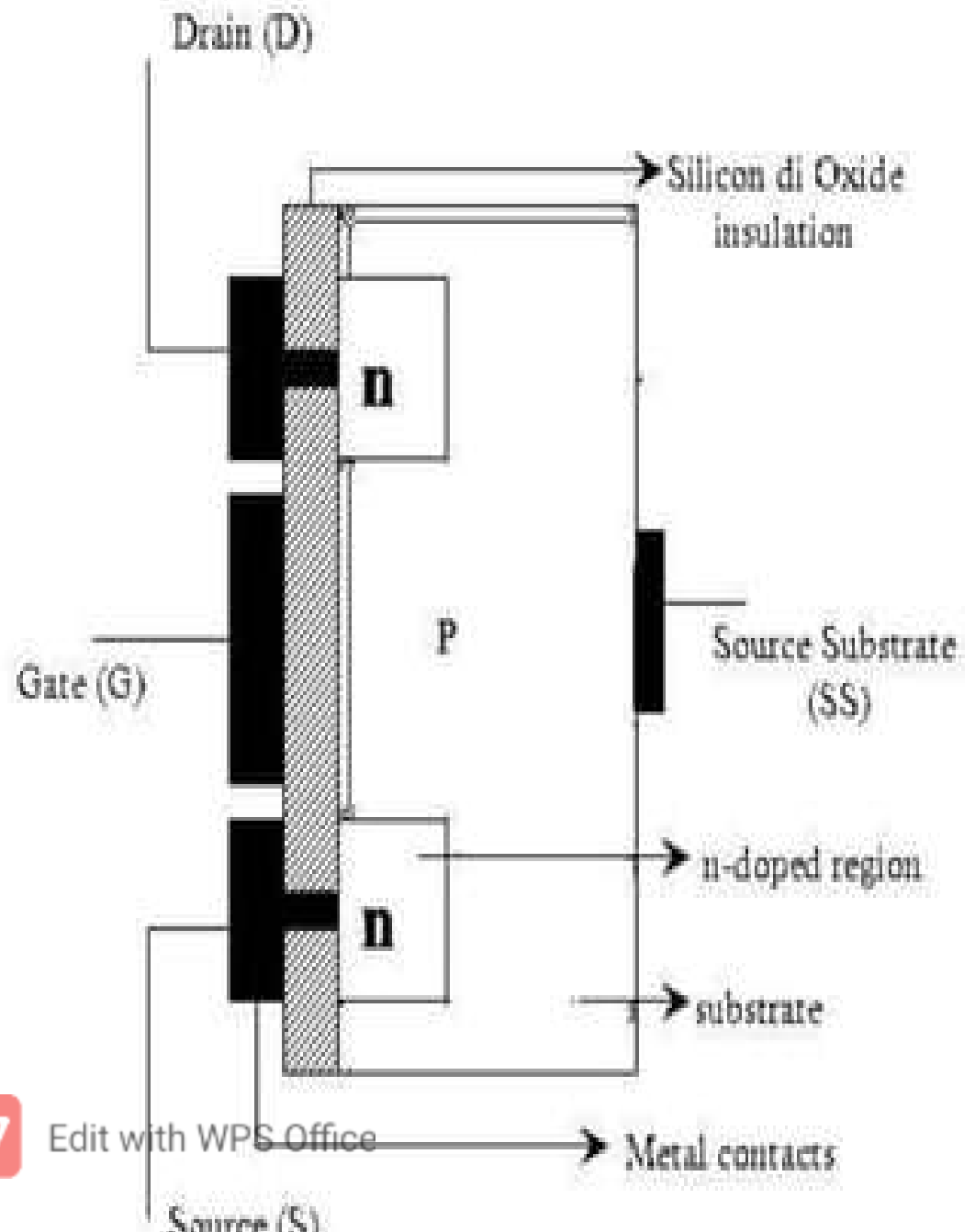
- Now on applying V_{DS} keeping drain at high potential, drain current will flow
- On increasing the value of V_{DS} , drain current will increase
- V_{DS} at which channel is pinched-off and drain current stop rising is called pinch-off voltage or saturation voltage
- At pinch of voltage drain current is called saturation current
- Now to increase saturation current further increase V_G



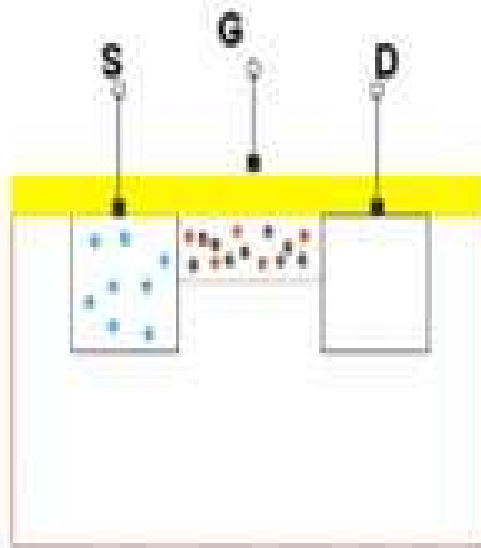
N-channel Enhancement MOSFET V-I Characteristics



N-channel Depletion MOSFET



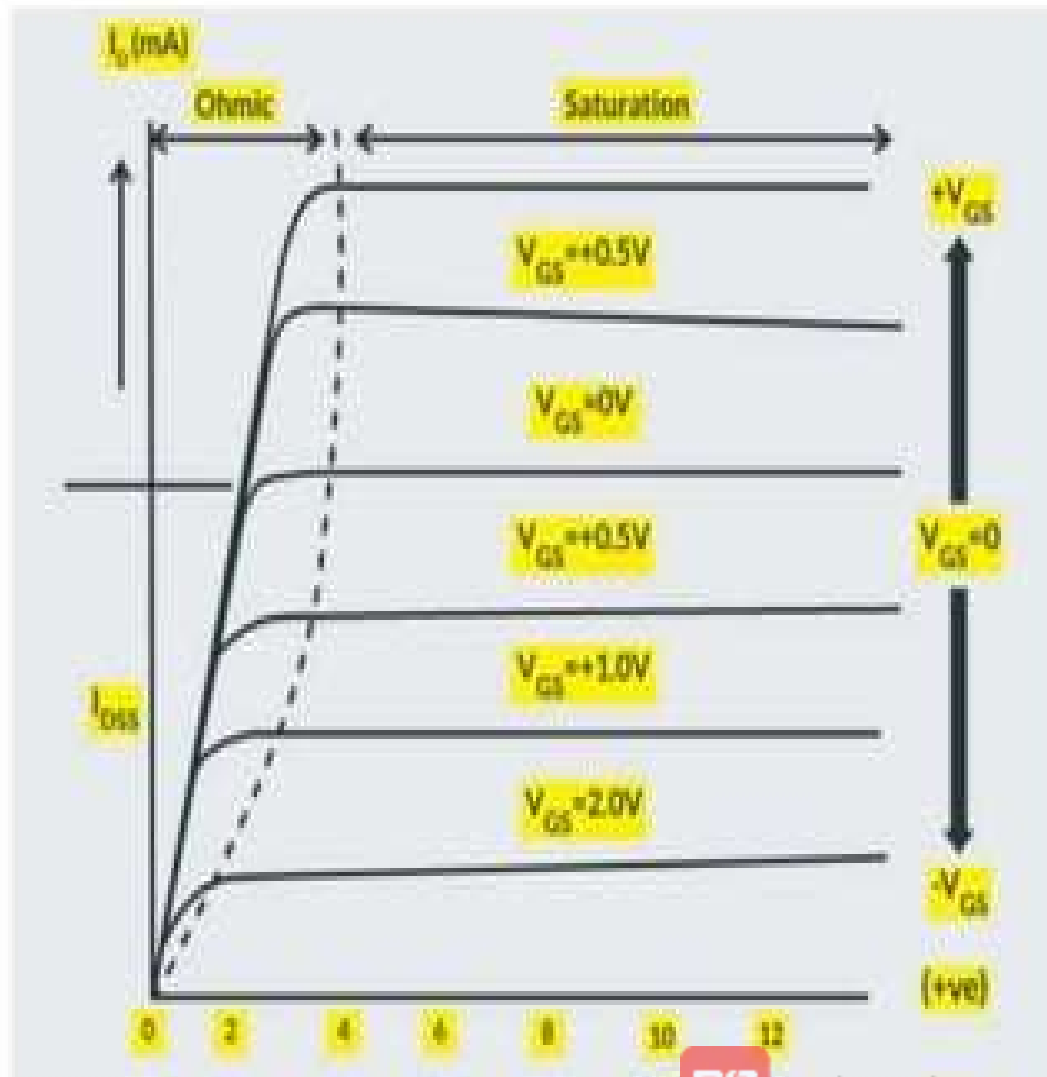
N-channel Depletion MOSFET



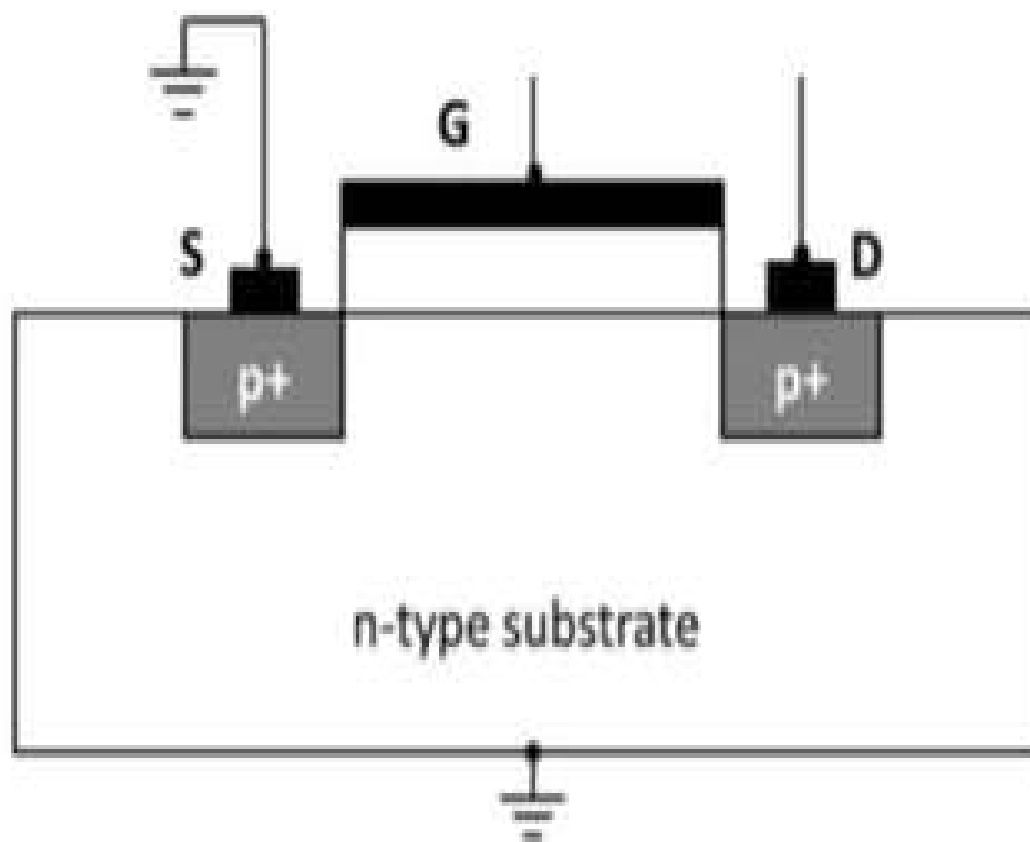
- N-type Channel is already present
- Thickness of channel increases on applying V_{GS} keeping gate at higher potential
- Now on applying V_{DS} keeping drain at high potential, drain current will flow
- On increasing the value of V_{DS} , drain current will increase
- On increasing V_{DS} further the N-channel towards drain will become narrow thus decreasing drain current
- V_{DS} at which channel is pinched-off and drain current stop rising is called pinch-off voltage or saturation voltage
- At pinch of voltage drain current is called saturation current
- Now to increase saturation current further increase V_{GS}



N-channel Depletion MOSFET V-I Characteristics



Operation Mode	Type	Condition	Equation
Cut-Off	NMOS	$V_{GS} < V_{TH}$	$I_D = 0$
	PMOS	$ V_{GS} < V_{TH} $	$I_D = 0$
Linear	NMOS	$V_{GS} \geq V_{TH}$ $V_{DS} \leq V_{GS} - V_{TH}$	$I_D = K_n(2(V_{GS} - V_{TH}) V_{DS} - V_{DS}^2)$
	PMOS	$ V_{GS} \geq V_{TH} $ $ V_{DS} \leq V_{GS} - V_{TH} $	$I_D = K_p(2(V_{GS} - V_{TH}) V_{DS} - V_{DS}^2)$
Saturation	NMOS	$V_{GS} \geq V_{TH}$ $V_{DS} \geq V_{GS} - V_{TH}$	$I_D = K_n(V_{GS} - V_{TH})^2(1 - \lambda V_{DS})$
	PMOS	$ V_{GS} \geq V_{TH} $ $ V_{DS} \geq V_{GS} - V_{TH} $	$I_D = K_p(V_{GS} - V_{TH})^2(1 - \lambda V_{DS})$



P-channel Enhancement MOSFET



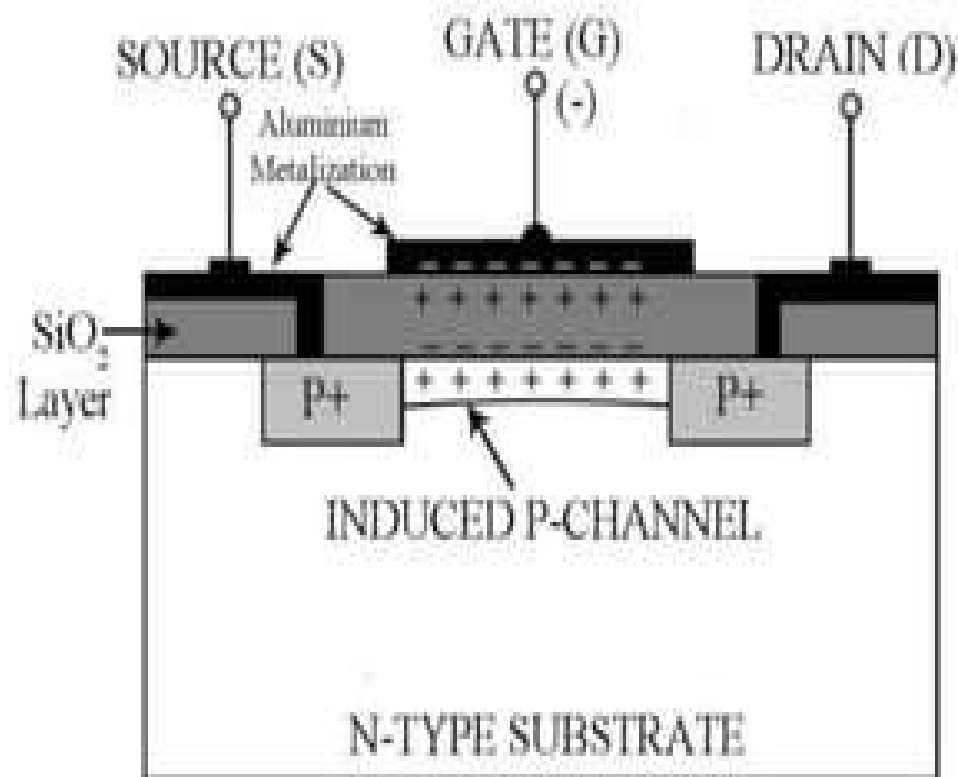
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P-channel Enhancement MOSFET

- It has N-type semiconductor material as body/substrate
- Two P-type wells are diffused on the body/substrate
- Two PN junctions are formed creating 2 depletion regions
- A metal contact is created at the bottom and a terminal is taken out known as body/ substrate terminal
- Same metal contacts are created on the top of P-type making source and drain terminals
- There is a layer of Silicon Dioxide in between the 2 P-type well which act as a dielectric
- A metal contact is created on the top of SiO₂ layer and a terminal is taken out called Gate
- Generally Source and body terminal are internally connected and is grounded

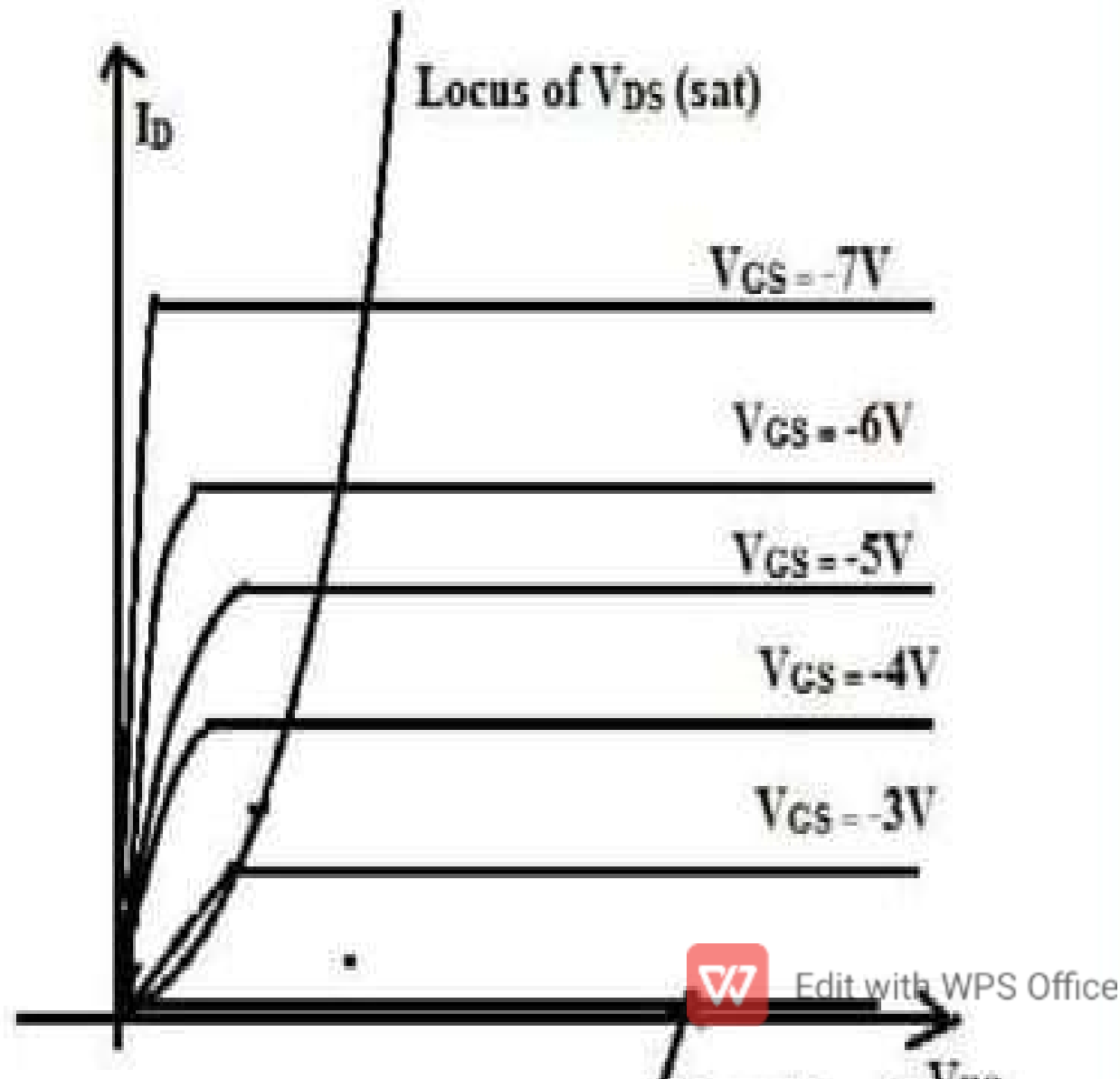


P-channel Enhancement MOSFET

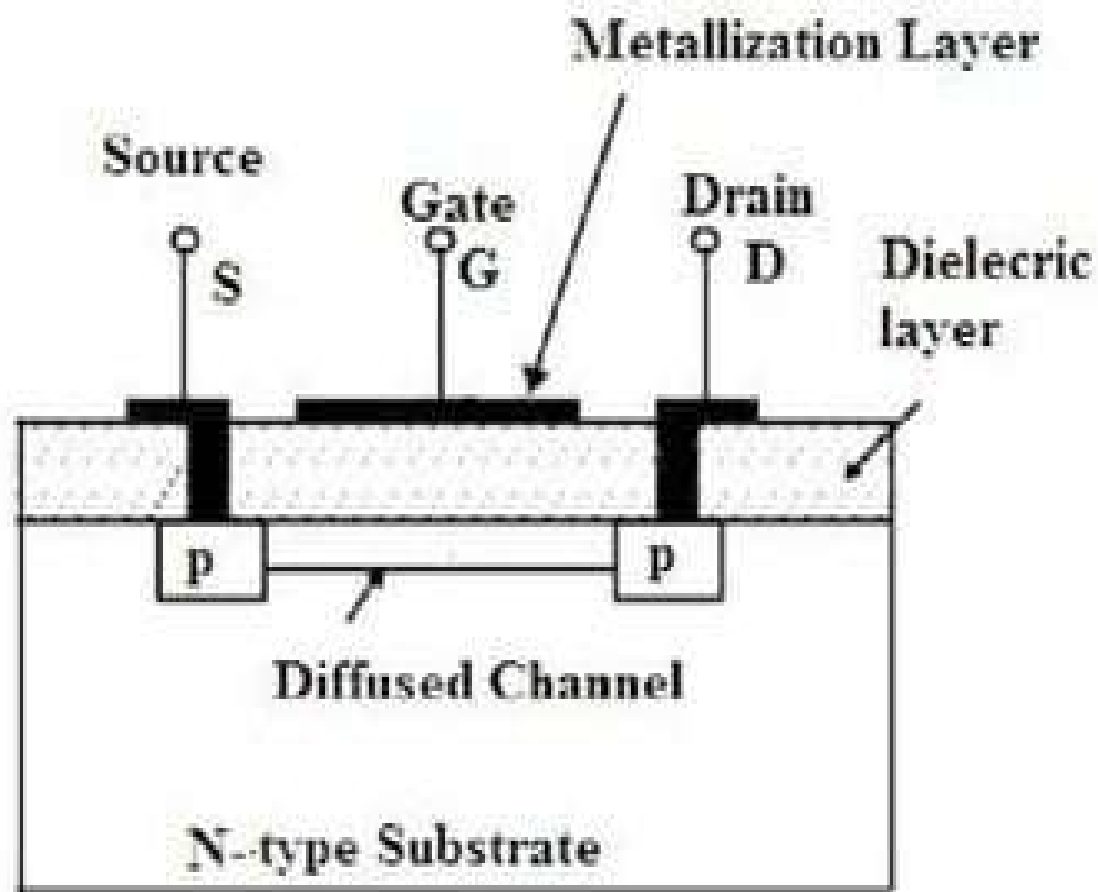


- Apply V_{GS} between gate and source terminal keeping gate at low potential
- Holes will push towards the gate terminal and accumulate there
- When V_{GS} reaches a threshold voltage, an p type channel is formed
- Now on applying V_{DS} keeping drain at low potential, drain current will flow
- On increasing the value of V_{DS} , drain current will increase
- V_{DS} at which channel is pinched-off and drain current stop rising is called pinch-off voltage or saturation voltage
- At pinch of voltage drain current is called saturation current
- Now to increase saturation current further make V_{GS} more





P-channel Enhancement MOSFET V-I Characteristics

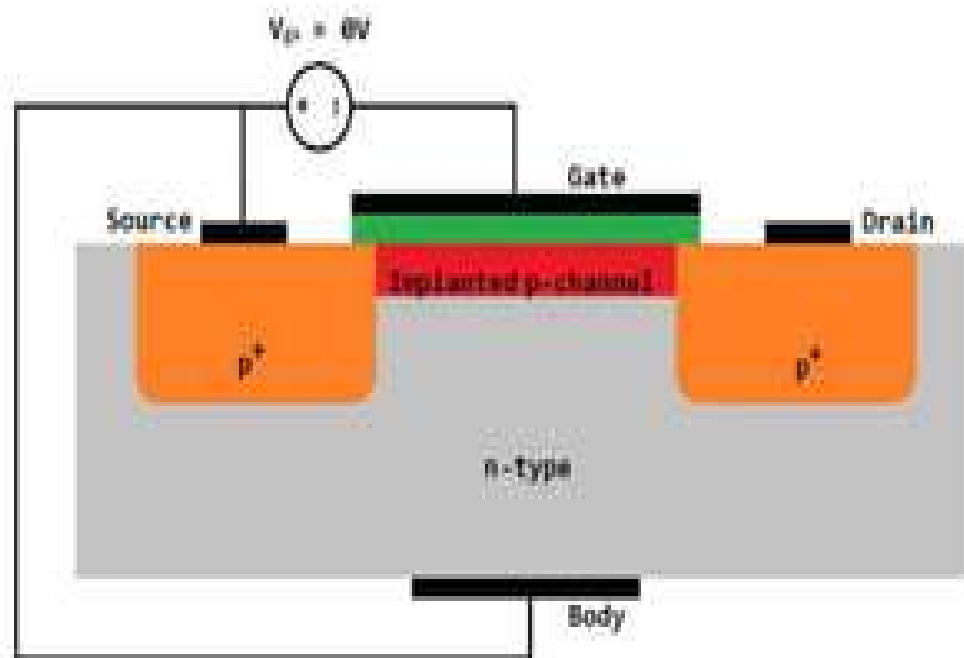


P-channel Depletion MOSFET



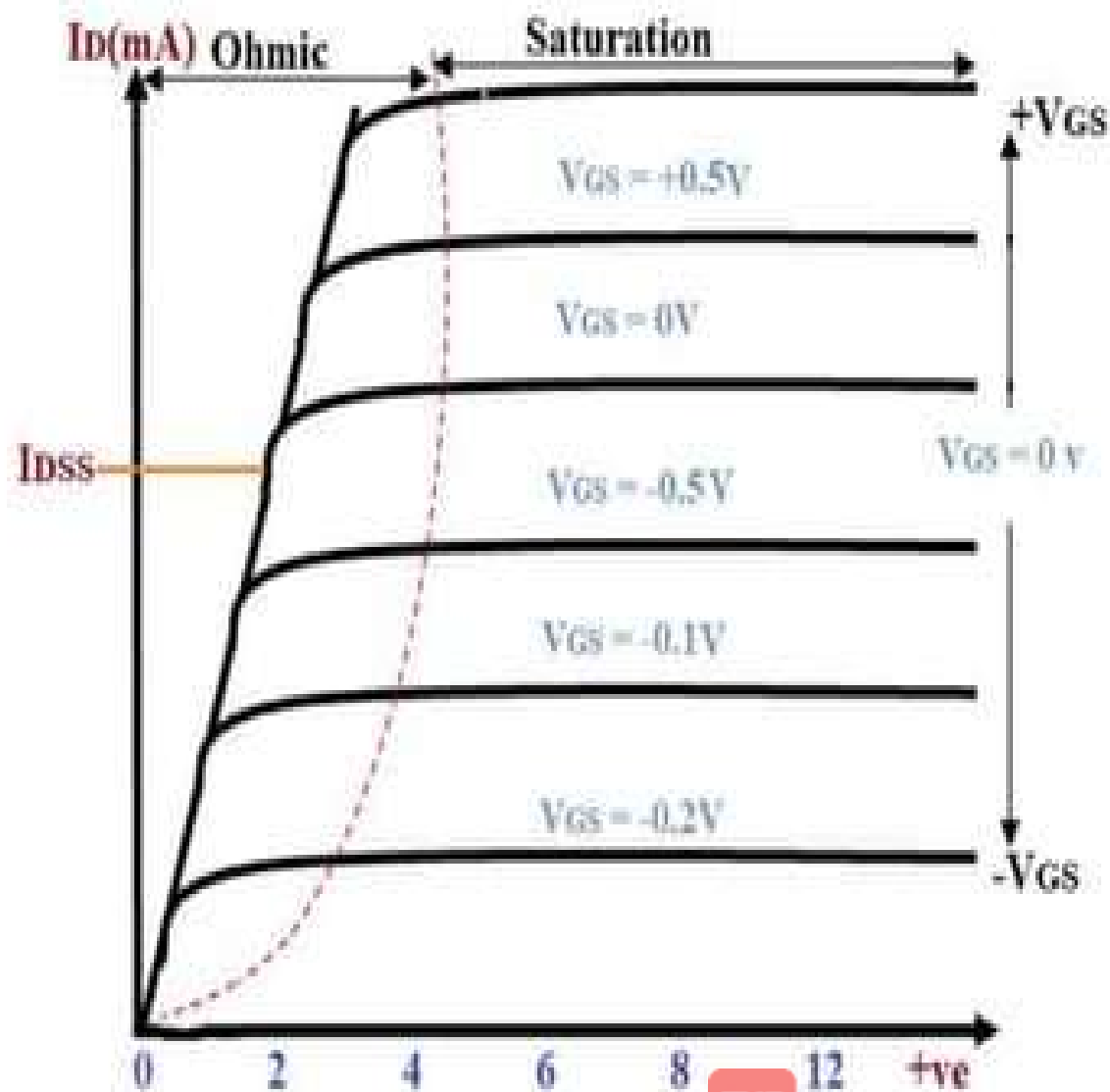
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P-channel Depletion MOSFET



- P-type Channel is already present
- Apply V_{DS} keeping drain at low potential, drain current will flow
- On increasing the value of V_{DS} , drain current will increase
- On increasing V_{DS} further the P-channel towards drain will become narrow thus decreasing drain current
- V_{DS} at which channel is pinched-off and drain current stop rising is called pinch-off voltage or saturation voltage
- At pinch of voltage drain current is called saturation current
- Now to increase saturation current further increase V_{GS}





P-channel Depletion MOSFET V-I Characteristics

